5 A

Gradient of P versus t graph = net force acting on the object.

$$= \frac{2.7 - 1.5}{0.4} = 3.0 \text{ N}$$

Since the momentum is decreasing, net force is acting opposite to motion

Hence, $5.0 - F = 3.0 \Rightarrow F = 2.0$ N

6 D